

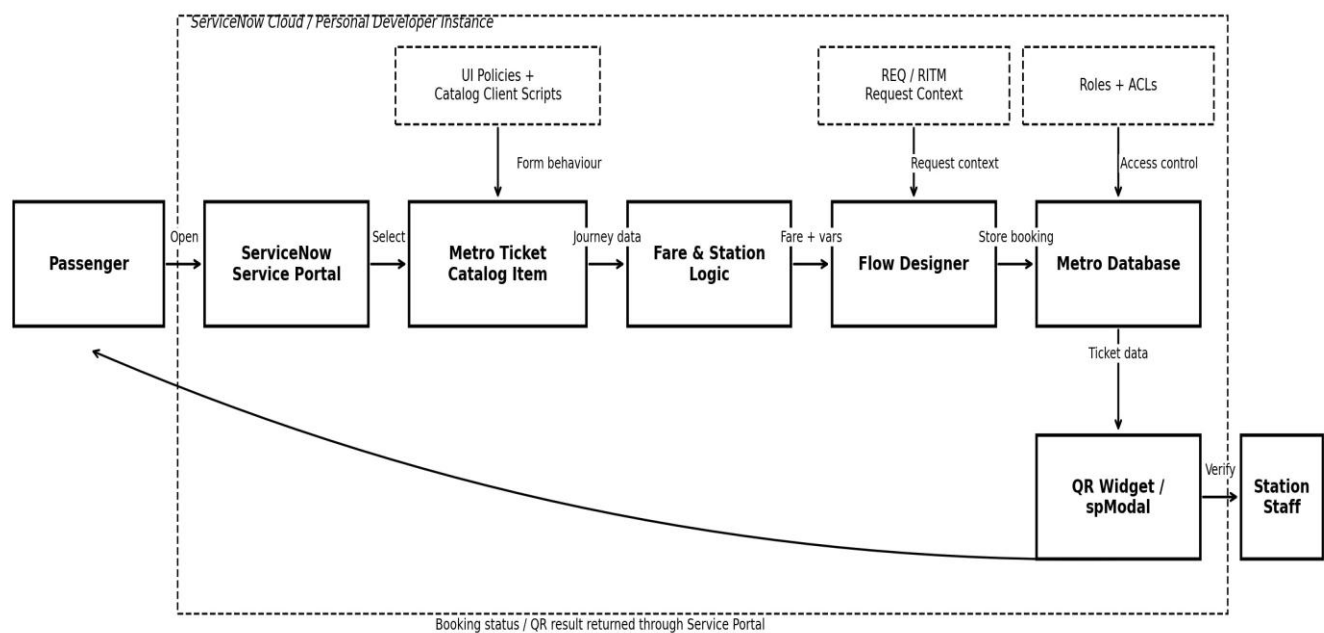
Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	29 September 2026
Project Name	Metro Ticket Generating System in ServiceNow
Maximum Marks	4 Marks

Technical Architecture:

The architecture is centered on the ServiceNow cloud platform. The passenger interacts through Service Portal and a Metro Ticket Service Catalog item. UI Policies and limited Catalog Client Scripts control the form experience; fare/station logic determines the ticket amount; Flow Designer orchestrates the booking; a custom Metro Database table stores structured ticket data; and a portal-compatible widget/spModal path presents the QR ticket. Authorized station staff and metro operations users validate and manage booking information.



Metro Ticket Generating System - Technical Architecture

Infrastructure Demarcation

- Cloud / Platform: ServiceNow instance (Personal Developer Instance during development/training).
- Client side: Web browser accessing Service Portal; no separate native mobile application is required for the documented solution.
- Application logic: ServiceNow configuration and low-code automation, primarily Flow Designer, UI Policies, catalog variables and limited Catalog Client Scripts.
- Data storage: ServiceNow request records plus a custom Metro Database table.
- External interfaces: No production payment gateway, SMS, WhatsApp or third-party API is claimed as implemented in the current version.
- Machine learning: Not applicable to the current project.

Table-1: Components & Technologies

S.No	Component	Description	Technology
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1	User Interface	Passenger-facing booking, request tracking and digital ticket experience.	ServiceNow Service Portal; Service Catalog; catalog variables
2	Application Logic-1	Dynamic form behaviour and essential client-side population/validation.	ServiceNow UI Policies; Catalog Client Scripts (minimal use)
3	Application Logic-2	Fare calculation and station mapping used to determine journey fare.	Configured ServiceNow logic / scripts used in the project; station/fare mapping
4	Application Logic-3	End-to-end request processing and automation.	ServiceNow Flow Designer; conditions; actions; dynamic task creation
5	Ticket Presentation	Portal-compatible display of the digital ticket / QR result.	Service Portal widget architecture; spModal; QR rendering logic
6	Database	Structured ticket / booking information.	ServiceNow custom table: Metro Database
7	Platform Records	Native request context created from Service Catalog submission.	ServiceNow Request / Requested Item (REQ / RITM) records
8	Notifications	Configured request / workflow notifications where used.	ServiceNow Notifications / Flow Designer notification actions
9	Security	Role-based protection of records and operations.	ServiceNow Roles and Access Control Lists (ACLs)
10	Diagnostics	Trace automation, variable values and failures.	Flow Designer Execution Details; platform logs / debugging outputs used during development
11	File Storage	No independent application file-storage service is required by the documented core flow.	ServiceNow-managed platform storage as applicable; no separate storage service specified
12	External API / Payment	Future integration point for payments or messaging.	Not implemented in current version; future UPI/payment/SMS/email/WhatsApp integrations
13	Machine Learning Model	No ML inference or model dependency is part of the current solution.	Not applicable
14	Infrastructure (Server / Cloud)	Application deployment and execution environment.	ServiceNow cloud instance / Personal Developer Instance for development

Technology Flow

Browser -> Service Portal -> Metro Ticket Catalog Item -> UI Policies / Catalog Client Scripts -> Fare & Station Logic -> Flow Designer -> REQ/RITM + Metro Database -> QR Widget/spModal -> Passenger / Station Validation.

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology / Design Choice
1	Open-Source Frameworks	No separate open-source application framework is required by the documented core implementation. QR generation may use code/library logic, but the supplied project description does not identify a specific library.	ServiceNow platform components; specific external QR library not asserted
2	Security Implementations	Protect booking records and administrative capabilities according to user role; enforce controlled read/write access.	ServiceNow roles, ACLs, platform authentication/session controls

3	Scalable Architecture	Configuration-driven design permits more stations, fare mappings, passenger types and ticket records to be added without replacing the complete solution.	ServiceNow multi-tier SaaS platform + modular Service Catalog / Flow Designer / custom table design
4	Availability	Application availability follows the availability of the deployed ServiceNow instance and related platform services. The project does not introduce a separate availability layer.	ServiceNow-managed cloud platform; no custom load balancer or distributed server configured by the project
5	Performance	Minimize unnecessary client scripting and use Flow Designer for maintainable automation. Test fare calculation, variable mapping and QR rendering for normal booking use cases.	ServiceNow server-side workflow execution + lightweight portal/client configuration
6	Maintainability	Use low-code/configuration-first components and document setup so another administrator can reproduce or modify the solution.	Service Catalog, Flow Designer, UI Policies, ACLs, documented configuration sequence
7	Observability / Troubleshooting	Trace workflow runs, condition paths and mappings when failures occur.	Flow Designer Execution Details; platform logs/debug outputs
8	Compatibility	Ensure the QR/ticket experience works in Service Portal rather than relying on backend-only UI behaviour.	Service Portal widget + spModalcompatible presentation

Implemented vs Future Technology Boundary

- Implemented/documented core: Service Portal, Service Catalog, catalog variables, UI Policies, limited Catalog Client Scripts, fare/station logic, Flow Designer, custom Metro Database table, REQ/RITM context, QR display via widget/spModal, ACL/role configuration, testing and execution tracing.
- Future integrations: payment gateway/UPI, SMS ticketing, email ticket delivery, WhatsApp booking/delivery and travel analytics dashboard.
- Not applicable in the current version: machine-learning model and a separately managed application server/database outside ServiceNow.